

## 科技英语常识题目整理

### Part1: understanding scientific literature

#### Ch1: Introduction of scientific literature

1. According to the form, scientific literature can be roughly classified into the following types:

- 1) journals; 2) monographs; 3) conference documents; 4) graduation papers; 5) technical reports; 6) science news; 7) patents; 8) product specifications; 9) government publications; and 10) standards.

常见迷惑项: Movie comments

2. According to the processing depth, scientific literature can be classified into:

(必考)

1) **primary literature**; 2) **secondary literature**; and 3) **tertiary literature**.

- Primary literature includes the original scientific materials published for the first time such as the papers that present novel scientific research, conference documents, patents, technical reports, and standards. 补充一个: graduation papers
- Secondary literature summarizes and synthesizes the primary literature, which includes **review articles** and **books** (eg: 论文集).
- Tertiary literature might consist of **textbooks**, **encyclopedias**, and similar works intended for **broad public consumption** (科普类的科学杂志).

3. the stylistic feature of SL (Scientific Literature) 格式特征

- Formality
- Conciseness
- Directness
- Objectivity
- Use of non-verbal language

4. the morphological feature of SL (Scientific Literature) 形态特征

(1) technical and scientific jargon TSJ

- general TSJ: can be used in different fields with different meanings
- specific TSJ: usually restricted to given fields

(2) Nominalization 名词化

- lexical nominalization 词汇名词化

- compounding 复合
- suffixation 后缀
- conversion 转变---动词具有名词含义

- clausal nominalization 从句名词化

- nominal clause
- gerund phrases 动名词
- infinitive phrase 不定式

### (3) Hedging 推诿

使用推诿的原因:

- 1) to minimize the possibility of another scholar opposing the claims;
- 2) to enable the results to be reported more precisely,
- 3) to enable a politeness strategy to be executed in case there may be flaws in the claims;
- and 4) to conform to an accepted practice in the field of academic writing.

Modal verbs, certain lexical verbs, certain modal adjectives, certain adverbs, certain nouns, introductory phrases

情态动词、特定实义动词、特定情态形容词、特定副词、特定名词、引导性短语

### 5. the syntactic feature of SL (Scientific Literature) 句法特征

- use of different tense
- use of the passive voice
- use of the subjunctive mood 虚拟语气
- use of imperative sentences 祈使句

## Ch2 Journals

1. 常考 **The purpose of a scientific journal** is to give researchers a venue to (1)**impart their knowledge to one another**, and (2)**make contributions to enriching the knowledge of all sciences**.

(3) to disseminate academic information 传播学术信息

(4) to lead the research directions

(5) to cultivate talents

常见的诱惑项:

- A. To assist researcher to get promoted quickly
- C. To help researchers to get more financial support
- D. To cultivate pupils to become great scientists

### 2. Types of journal articles

A. Original research articles

B. Review articles/survey articles/overview articles

(考点) A review article summarizes the current state of understanding of a topic

C. Perspective, opinion or commentary articles 观点、评述类文章

However, the author needs to have an in-depth knowledge of the topic.

D. Book reviews 书评

(可能考点) A book review is a descriptive and critical/evaluative account of a book.

作用:

provides a summary of the contents

assesses the value of the book

recommends it (or not) to other potential readers

### 3. Journal metrics

#### 目的（常考）：

- A. Reflect the place of a journal within its field,
- B. the relative difficulty of an article published in that journal,
- C. and the prestige associated with it.

有哪些？

#### 01 IF or JIF 影响因子

The Impact Factor (IF) or Journal Impact Factor (JIF) of an academic journal is a measure reflecting the yearly average number of citations to recent articles published in that journal. 年平均引用量

#### 02 Eigenfactor 特征因子

Journals are rated according to the number of incoming citations, with citations from highly ranked journals weighted to make a larger contribution to the Eigenfactor than those from poorly ranked journals.

期刊依据 incoming citations（引用量）进行评级；高排名期刊的引用权重更高，其对特征因子（Eigenfactor）的贡献大于低排名期刊的引用。

#### 05 h-index（考点）

The h-index is an **author-level metric** that attempts to measure both the productivity and citation impact of the publications of a scientist or scholar.

反映学者成就 至多 h 篇论文分别引用了至少 h 次

#### 08 Immediacy Index 立即索引

The Immediacy Index is the number of citations the articles in a journal receive in a given year divided by the number of articles published.

某一年该期刊全部论文获得的总引用量 ÷ 该期刊当年刊发论文总数

#### 09 Total Cites

Total Cites refer to the total number of times that a journal has been cited by all journals included in the database in the Journal Citation Reports (JCR) year.

《期刊引证报告》（JCR）统计年度内，该期刊被数据库中全部期刊引用的总次数。

#### 10 Cited Half-Life 引用半衰期

Cited Half-Life measures the number of years, going back from the current year, that accounts for half the total citations received by the cited journal in the current year.

从当前年份向前回溯，累计被引次数达到该期刊当年总被引量一半所对应的年份跨度。

- **数值越大**：期刊多年前的旧论文至今仍在被大量引用，文献具备长期、持久的学术价值，经典成果生命力强；  
典型：文史、基础理论类期刊，半衰期普遍 5 年以上。
- **数值越小**：只有近 1-2 年的新论文才有引用热度，旧文献很快失去关注度，领域更新迭代极快；  
典型：计算机、生物医药、前沿工程类期刊，半衰期多 1-3 年。

#### 4. Identification code

The identification code of a journal is a specific code for identifying a particular journal.

##### 01 DOI

Digital Object Identifier (DOI)：数字对象标识符

##### 02 IDS Number

The ISI Document Solution (IDS) Number uniquely identifies the journal and issue.  
用来标识期刊

##### 03 ISSN

An International Standard Serial Number (ISSN) is an **8-digit serial number** used to identify a serial publication, and is especially helpful in distinguishing between series with the same title.

国际标准连续出版物号（ISSN）是一组 8 位数字编号

##### 04 ISBN（考点）

The International Standard Book Number 国际标准书号

**2007.1.1 之前是 10 位，之后是 13 位**

#### 5. Citation index 引文索引

A citation index is a kind of **bibliographic index**（文献索引），an index of citations between publications, allowing users to easily establish which later documents cite which earlier documents.

| Natural science                                              | Social science                                                                        | China |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------|-------|
| SCI                                                          | SSCI                                                                                  | CSCD  |
| EI                                                           | A&HCI                                                                                 | CSSCI |
| CPCI-S<br>Conference Proceedings<br>Citation Index - Science | CPCI-SSH<br>Conference Proceedings<br>Citation Index - Social<br>Science & Humanities |       |
| ISR<br>Index to Scientific<br>Reviews 科学评论索引                 |                                                                                       |       |

SCI:The Science Citation Index (SCI) is a citation index originally produced by the **Institute for Scientific Information (ISI)** and created by **Eugene Garfield**.

EI: Ei Compendex is the broadest and most complete **engineering literature** database available in the world. (考过)

ISTP: The Index to Scientific & Technical Proceedings (ISTP) is a scholarly literature database established in 1978.

A&HCI: The Arts & Humanities Citation Index (A & HCI)

JCR: developed by **Clarivate Analytics**, provides journal intelligence that highlights the value and contribution of a journal through a rich array of transparent data, metrics, and analysis.

### Ch3 Monographs 专著

A monograph is a specialist work of writing on a single subject or an aspect of a subject, sometimes called a scholarly treatise.

特别的点: Usually, only one author is involved, but the author may collaborate if necessary

There are general and scholarly monographs.

1. There are **some standards** that can be used to measure if a book fits the description of a scholarly monograph.

- 1) The author has an advanced degree (e.g. a Ph.D. or Ed.D.), and/or is associated with an academic institution
- 2) The book is published by a university press, scholarly society, or other scholarly organizations.
- 3) The book includes a bibliography at the end citing all of the sources that the author used in the research.

### 2. Features of scholarly monographs

- (1) Academicity
- (2) Creativity: The most significant feature
- (3) Authoritativeness 权威性
- (4) Comprehensiveness
- (5) Systematicness 系统性
- (6) Beyond time and space 跨时空性

### 3. Differences between academic articles and monographs

- (1) the length of words

学术文章 4000-5000

专著 40000-50000

- (2) structure:

学术文章: general structure+focusing on one argument

专著: follow a structure of its own+focusing on one specialty and giving thought to all relevant issues

(3) content 考过:

学术文章: 集中解决一个问题

专著: 解决多个问题

an academic article proposes a **hypothesis** and proves it with evidence, while a monograph conducts **comprehensive and intensive illustrations** of more issues.

Ch4 Conference documents

1. Types of conference documents (考点)

According to conference stages, they can be classified into three main types:

1) Documents distributed before a conference.

This kind of documents includes announcements, calls for papers, instructions for authors, instructions for speakers, registration forms (报名表), invitations, preliminary printed proceedings (初步议程), and collections of abstracts.

2) Documents distributed at a conference.

This kind of documents includes daily schedules, conference proceedings 【such as conference paper abstracts, opening speeches, and closing speeches.】

3) Documents distributed after a conference.

This kind of documents includes post-conference proceedings, special editions of conference documents, conference reports, transactions(议事录), and conference promotions.

2. Types of academic conferences

A. Conference

B. Forum: public

C. Lecture

D. Panel 专家咨询会

E. Seminar 研讨课

F. Symposium 主题研讨课

G. Workshop (考过) 工作坊

A workshop is a usually **brief educational program** for a relatively small group of people that focuses especially **on techniques and skills in a particular field**. It can be a small session held during conferences.

Ch5 Graduation papers

1. Types

Graduation papers can be classified into different types from different perspectives. According to the degree, it can be classified into the graduation paper for a bachelor's degree, the graduation paper for a master's degree, and the graduation paper for a doctorate.

According to the research method, it can be classified into the theoretical research paper, the experimental research paper, and the descriptive research paper.

According to the research field, it can be classified into the research paper in arts and humanities and the research paper in science and engineering.

2. Differences between a graduation paper and a journal article

In terms of **purpose**, a graduation paper is to justify that the student has reached a certain academic level in his or her specialty and can graduate with a certain level of academic degree. A journal article, however, is to indicate that the author has accomplished a research project and wants to share his or her research results with others.

In terms of **length**, a graduation paper is much longer than a journal article due to the word number requirements.

#### 考点

**In terms of the number of authors, a graduation paper is written by only one author, while a journal article can be co-written by more than one author.**

### Ch6 Technical reports

Technical reports are often prepared for **the sponsors of research projects** and written according to general standards, corporate design standards of the sponsors, logical rules, and practical experiences. They are considered "**non-archival**" publications and free to be published elsewhere in peer-reviewed venues with or without modifications. 可自行将文稿修改或不做修改，另行发表于其他同行评审出版物。

#### 1. Types 涉及考题

(1) In terms of **content**, technical reports can be categorized into research findings reports, technical information reports, technical specifications reports 技术规范, technical evaluation reports, technical recommendation reports, and technical manuals and instructions reports 技术手册和操作说明报告.

(2) In terms of **research progress**, technical reports can be categorized into primary reports, progressive reports 进度报告, interim reports, and final reports.

(3) In terms of the **scope of circulation**, technical reports can be categorized into top secret reports, secret reports, confidential reports, restricted reports, unclassified 不保密的 reports, and declassified 撤销保密 reports.

(4) In terms of formality, technical reports can be categorized into informal reports, semiformal reports, and formal reports.

#### 2. Features 可能考察

As a special kind of scientific literature, technical reports usually **(1)report edge cutting technologies, (2)tend to be confidential 涉密性, and(3) can be found through their serial number 序列号.**

The serial number is the most important identification code for the report.

#### 3. Four series of technical reports in America 可能考察

##### A. AD reports

AD (ASTIA Document) was originally provided by the Armed Services Technical Information Agency (ASTIA) in 1951.

Most of the technical reports in the AD and AD-A series can be available to the public. But those in the other more restricted series (AD-B, AD-C, AD-D, etc.) available to

the public are far fewer.

#### B. PB reports

In 1945, in order to acquire and disseminate the vast amounts of scientific and technical information created as a result of federal contract research during and after World War II, the Publication Board (PB) was established in America.

Today, the PB series comprising U.S. government issued reports continues to be distributed by the Department of Commerce and collected by other institutions as well as the Library of Congress 国会图书馆.

#### C. NASA reports

NASA reports refer to all scientific and technical documents provided by the National Aeronautics and Space Administration (NASA), an independent agency of the Federal Government.

#### D. DOE reports

Initially reports of the Atomic Energy Commission (AEC) 美国原子能委员会, this series expanded and diversified along with the agency when it became the Department of Energy (DOE).

### Ch7 Science news

#### 1. 一些概要

定义: Science news is a factual report on recent happenings in the field of science and technology.

特点: Science news has the nature of both **news and science**

It is **new** in terms of content and time. It is supposed to disseminate scientific knowledge, so it should be **presented accurately and based on scientific evidence**.

Science news is A.explicit 清晰的, B. precise, and C. tries not to include jargon.

Science news is explicit, precise, and tries not to include jargon. The subject-verb-object sentence structure is often used; vivid and lively expressions are often employed; and anecdotes, examples and metaphors are often offered. Moreover, background information is often provided to facilitate understanding on the part of readers.

#### 2. Types

A. Science news stories 科技新闻报道

timely scientific events of interest and significance to readers

B. Science newsletters 简讯

C. Science feature stories 专题报道

D. Science news commentaries

### Ch8 Patents 专利

#### 1. 定义 (经常考)

**A patent is an exclusive right 排他性的权利 granted for the protection of an invention.**

特点:

- A. It provides its owner the exclusive right over the exploitation (不正当利用) of the invention.
- B. Thus, in general, the patent owner can prevent others from making, using, selling, or importing the patented invention without his or her permission.
- C. It is granted by the government to the inventor for a period of usually **20 years** from the filing date 存档 of a patent application.
- D. Patents are territorial 地域性的

## 2. Types

- A. Invention: a new technical solution worked out to a product, to a method or to the improvement of the product or the method
- B. Utility models: a new technical solution applicable to the shape, structure, or both, of a product.
- C. Design: a new design of the shape, structure, or both, of a product, which gives an aesthetic feeling and has industrial applications

## 3. Copyright 版权

(1) Copyright, like patent, is also a form of intellectual property. But it mainly grants the creator of an original work exclusive rights for its use and distribution, usually for a limited time.

(2) It is often shared among **multiple authors**, each of whom holds a set of rights to use or license the work.

(3) Copyrights are considered territorial rights

(4) Typically, the duration of a copyright spans the author's life plus 50 to 100 years  
考过

① In the case of **joint works**, copyright protection is granted for the length of the life of the last surviving joint creator plus another 70 years.

② Works made for hire 职务, as well as anonymous 匿名 and pseudonymous 笔名 works, are protected for a term of either 95 years from the year of first publication 自首次出版之日起 95 年 or 120 years from the year of creation 自创作完成之日起 120 年, whichever is shorter 以二者中期限较短者为准. When the term of protection for a copyrighted work expires, the work enters into the public domain.

当受版权保护作品的保护期届满后, 该作品即进入公有领域

## Ch9 Product specifications 产品说明

### 1. 定义 definition

Product specifications are documents that provide defining information about a product. They include the manufacturer; a list of rules, bans, and standards that apply to the item; design specifications and product images that visually illustrate the product.

- 2. Features
  - A. Genuine
  - B. Scientific
  - C. Logical
  - D. Pratical

### 3. Types 经常考

In terms of fields, product specifications can be classified into:

- 1) industrial product specifications;
- 2) agricultural product specifications;
- 3) financial product specifications;
- 4) insurance product specifications. 保险

## Ch10 Government publications 政府出版物

### 1. 定义 (考过)

Government publications, or official publications, are documents and information produced by governments at different levels. They are intended to make information known to the public in an **authoritative** and **open** way, so they are authoritative and communal 公共的.

### 2. International government publications (IGP) 了解即可

#### A. UN publication

*Important publications of the UN include Yearbook of the United Nations, International Court of Justice Yearbook, The United Nations Disarmament Yearbook, UNCTAD Handbook of Statistics, UN Chronicle, etc.*

#### B. OECD publication

As an intergovernmental economic organization officially born in 1961 to stimulate economic progress and world trade, the Organisation for Economic Co-operation and Development (OECD) is committed to the market economy, providing a platform to compare policy experiences, seek answers to common problems, identify good practices, and coordinate domestic and international policies of its members.

### 3. American government publications (没考过, 可能考)

#### A. GPO publications

The U.S. Government Publishing Office (GPO)

#### B. GPO monthly catalog 月报

#### C. NTIS publications

The National Technical Information Service (NTIS) is an agency within the U.S. Department of Commerce

It serves as the largest central resource for government-funded scientific, technical, engineering, and business-related information available today.

### Ch11 Standards

1. 定义: Standards refer to documents that A.provide requirements, B.specifications, C.guidelines, or D.characteristics that can be used consistently to ensure that materials, products, processes, and services are safe, reliable, and consistent.

Standards are actually of legal nature

### 2. Types(考点)

根据 applicable scope 划分

- A. International standards
- B. Regional standards
- C. National standards
- D. Professional standards 行业标准
- E. Institutional standards 机构标准

根据 内容 划分

- A. Basic standards
- B. Manufacturing standards
- C. Standards of method

### 3. ISO and ISO 9000 Standards

ISO is the International Organization for Standardization that sets standards in many businesses and technologies, including computing and communications.

ISO 9000 can be defined as a set of international standards on quality management and quality assurance developed to help companies effectively document the quality elements needed to maintain an efficient quality system. They are not specific to any one industry and can be applied to organizations of any size. ISO 9000 can help a company satisfy its customers, meet regulatory requirements, and achieve continual improvement. First published in 1987 by ISO, it has already been adopted by about 60 countries. It is considered the first step or the base level of a quality system.

## Part2:reading research articles

### Ch12 Title and author affiliation

1. 定义: The title is the main indication of an article's subject and the main source of information for doing relevant judgment at the time of literature searching.
2. 特点: The title of a research article is usually characterized by:
  - 1) using more nouns or noun phrases, or gerunds;
  - 2) being brief and concise;
  - 3) being specific;
  - 4) using more compounds.

#### 3. Types of titles ( 会给一个标题让你判断具体的类型 )

##### A. Declarative titles 陈述型标题

Eg:Subcutaneous adipose gene expression prior to weight loss can differentiate and weakly predict dietary responders 完整陈述句

##### B. Descriptive titles 描述型标题

Eg:A worldwide phylogeography for the human X chromosome 词组

##### C. Interrogateive titles 设问型标题

Eg:Are there rearrangement hotspots in the human genome? 问号

##### D. Compound titles 复合型标题

Eg:Estimating vadose zone hydraulic properties using ground penetrating radar: The impact of prior information 主副标题

#### 4. Author affiliation

- ① the lead author
- ② the corresponding author ( the one to be contacted if reader want)

Generally, the lead author is listed at the beginning, and the corresponding author is usually marked with a little star

### Ch13 Abstract

#### 1. 组成部分

background problem methods results conclusion

Additional part: objectives 研究目的 limitations

#### 2. Types

There are four common types of abstracts:

##### A. informative abstracts

It summarizes the key information from every major section of the article, and provides the key facts and conclusions.

##### B. indicative abstracts/descriptive abstract

merely includes information about the purpose, scope, and methods used in the study 更短

C. structured abstracts

consists of a few paragraphs, each preceded by a subheading 分段 + 子标题

D. graphical abstracts.

Ch14 Introduction

1. 组成部分

- A. Establishing a territory background information
- B. Reviewing previous related research literature review
- C. Establishing a niche research gap
- D. Occupying the niche thesis statement

Ch15 Methods

1. 组成部分

Generally, this section includes materials, instruments, subjects 被试者, procedures, data collection, and data analysis.

2. Ethical approval (出题点)

需要 ethical approval 的研究有哪些?

- 1) research involving living human subjects;
- 2) research involving **human remains 残骸**, cadavers 尸体, tissues, biological fluids 生物流体, embryos 胚胎, etc.;
- 3) research that involves gathering information about human beings and organizations through interviewing, surveying, questionnaires, observation of human behavior, audio/video taping, and administering tests or stimuli;
- 4) research involving secondary use of data ( use of data initially collected for another purpose)

Ch16 Results

1. 特点 (考点)

- A. in the form of figures and tables.
- B. preferably in a chronological and orderly sequence 逻辑顺序
- C. It does not have to interpret, explain, or analyze the research results.
- D. often the shortest section of a research article. But it is the most valuable section of the research article, which reveals the academic value of the article.

2. Features

- A. Objective
- B. Logical
- C. Concise
- D. non-verbal language---figures/tables
- E. Mixed tense

3. Moves (可能出题)

- A. Providing preparatory information
- B. Describing the data in figures and tables
- C. Reporting the results based on the data

#### 4. Figures and tables

01 bar chart

02 pie chart

03 flow chart 流程图

04 line chart

05 scatter plot 散点图

06 area chart

07 bubble diagram

08 structure schematic diagram 流程示意图

#### Ch17 Discussion

##### 1. 组成部分

- 1) answers to the proposed questions by summarizing the main significant findings of the research;
- 2) reasons and causes for the results and explanations for how the results support the answers and how the answers fit in with the existing knowledge on the topic;
- 3) implications of the research by providing the researcher's interpretations and opinions;
- 4) association and comparison of the findings with the existing research results;
- 5) limitations of the research and recommendations for further research.

##### 2. 特点

- A. It is organized from the specific to the general.
- B. It has the same point of view as that in the Introduction section.
- C. There are facts as well as opinions or comments in this section.

##### 3. Moves

- 1) Move 1-reviewing the present study;
- 2) Move 2-consolidating the results;
  - reporting the results
  - explaining the meaning of the findings and why they are important
  - relating the results to those of similar studies
  - providing alternative explanations for the results
- 3) Move 3-stating limitations and possible future research directions.

#### Ch18 conclusion

##### 1. Usually, this section contains the following main information: (考点)

- 1) restatement of the research question and thesis statement;
- 2) summary of the main points in the article;
- 3) discussion of the significance of the research;

4) presentation of the future of the subject.

## 2. Types

### A. Thesis-oriented conclusion

- introductory statement
- consolidation of the research gap
- recommendations

### B. Field-oriented conclusion

mainly focuses on the field and only mentions the thesis and its findings or contributions in the context of **the whole field**.

## Ch19 Acknowledgement

### 1. 考题：问致谢对象是否是正确的

Generally, the section of Acknowledgements **starts with the most important people**, followed by **the financial support** that has been received from a foundation or research group, such as a grant 补助金, a fellowship, or a scholarship. **Lab assistants, classmates, parents, friends, partners, or other acquaintances** can also be thanked in this section.

## 2. Features

- A. Polite
- B. Brief
- C. Specific

## Ch20 References

### 1. 参考文献字母代表含义

- A: 析出文献 (论文集) article
- M: 专著 monograph
- C: 论文集 collection
- N: 报纸文章 newspaper
- J: 期刊文章 journal
- D: 学位论文 dissertation
- R: 报告 report
- S: 标准 standard

2. Works cited = references 文中提到的 ≠ bibliography 文章提到的+有启发性的

### 3. Types

#### A. In-text references/citations 文内引 labeling

- ① Reference
- ② Quote 引用原文
- ③ Summary
- ④ Paraphrase of another source

分类:

- direct citation

Direct citation means citing the text **word by word** with quotation marks.

- indirect citation

Indirect citation means **paraphrasing or summarizing** the text from the original writing

B. End references/reference list 文末引

4. Citation styles

可能出题: 识别 styles

A. APA style

The American Psychological Association (APA) style is an **author/date-based** style.

In this style, the emphasis is placed on **the author and the publication date** of a piece of work.

In the APA style, an in-text citation consists of the author's last name and the date of publication. For example:

*During the outbreak, the doctors thought the psychological symptoms were unrelated (Juarez, 1993).*

If the author is named in the text, there will only appear the publication date in parentheses. For example:

*Juarez (1993) has analyzed many reports written by psychologists directly involved in the studies.*

When citing a work by two authors, the last names of both authors will appear in parentheses. Use an ampersand (&) to separate the names in the citation, but use the word "and" in the text. For example:

- ▶ *The small tribes along the Amazon that have survived over the centuries have evolved in parallel ways (Hanes & Roberts, 1978).*
- ▶ *Hanes and Roberts (1978) claim that the ways in which the small Amazonian tribes have evolved over the centuries are similar to each other.*

Sometimes a work by three to six authors is cited. In this case, the last names of all the authors are cited in the first citation. Then, in following citations, only the first author's last name appears followed by "et al.". For example:

- ▶ *Living on the road for weeks at a time has been linked to many negative*

*emotional, psychological, and physical health issues (Hans, Ludwig, Martin, & Varner, 1999).*

and then:

- ▶ *According to Hans et al. (1999), lack of stability is a major factor.*

If a work by seven or more authors is cited, the last name of the first author appears followed by "et al." in both first and subsequent citations. The complete list of authors is included in the works cited list at the end of the paper. For example:

*As Carnes et al. (2002) have noted, the immediate bond between a newborn baby and its mother has been extensively studied by many disciplines.*

B. MLA style

The Modern Language Association (MLA) style is most often **applied by the arts and humanities**, particularly in the U.S. It is arguably the most well used of all of the citation styles.

In the MLA style, in-text citations use the author's last name and if possible a page number within the source. For example:

*Romantic poetry is characterized by the "spontaneous overflow of powerful feelings"*  
(Wordsworth 263).

If the author is named in the text, only the page number is included in parentheses. For example:

*Wordsworth extensively explored the role of emotion in the creative process* (263).

When citing a work by two or three authors, the last names of the authors and the page number are put in parentheses. For example:

- ▶ *The historian's main job is to search for clues and solve mysteries* (Davidson and Lytle 6).
- ▶ *With the advent of behaviorism, psychology began a new phase of inquiry* (Cowen, Barbo, and Crum 31-34).

If the source has more than three authors, the first author's last name appears in parentheses followed by "et al.". For example:

*Helping each family reach its goals for healthy child development and overall family well-being was the primary approach of Project EAGLE* (Bartle et al. 35).

### C. IEEE style

The IEEE reference style is the standard referencing style set by the Institute of Electrical and Electronics Engineers (IEEE) and is based on the widely used Chicago referencing style.

A number enclosed in square brackets, such as [1] or [2, 6] or [1-6], is placed in the text to indicate the relevant reference. Citations are numbered in the order in which they appear in the text and each citation corresponds to the numbered reference appearing at the end. For example:

*Energy management is a growing concern for transportation systems [1-3]. Metro train systems (or, "the metro") have developed rapidly in China due to their convenience, rapidity, punctuality, safety, and reliability. To reduce the operational cost and ensure an environmentally-friendly mode, researchers have focused much attention on the energy consumption of the metro in recent years [4-6].*